

Companion LXV: Fréchet Means of Persistence Diagrams in the Relaxation-Driven Cyclic Cosmology

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Abstract

Fréchet means provide a statistical summary of persistence diagrams by minimizing the expected Wasserstein distance to an ensemble of diagrams. In weak-lensing analyses, they characterize the average topological structure of convergence fields across realizations, redshift bins, or smoothing scales. In the standard cosmological model, Fréchet means reflect the nonlinear matter distribution and the projection of large-scale structure. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the ensemble topology, shifting the mean diagram in a characteristic way. This Companion develops a continuous description of Fréchet means in RDCC, deriving how the relaxation scale influences ensemble topology, diagram geometry, and the observational signatures in weak-lensing surveys. The resulting framework provides a distinctive ensemble-level probe of the RDCC gravitational field.

1 Introduction

Persistence diagrams encode the birth and death of topological features in weak-lensing convergence fields. While Wasserstein distances compare individual diagrams, Fréchet means summarize entire ensembles by minimizing the expected distance to all diagrams in the set.

In the standard cosmological model, Fréchet means reflect the nonlinear matter distribution and the projection of large-scale structure. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the ensemble topology in a scale-dependent manner, producing a distinctive signature in Fréchet-mean analyses.

2 Fréchet Means of Persistence Diagrams

Given an ensemble of persistence diagrams $\{D_i\}$, the Fréchet mean D^* is defined as

$$D^* = \arg \min_D \sum_i W_p(D, D_i)^p, \quad (1)$$

where W_p is the p -Wasserstein distance.

The Fréchet mean represents the diagram that best summarizes the ensemble.

3 RDCC Modification of Persistence Diagrams

In RDCC, the convergence evolves as

$$\kappa_{\text{RDCC}}(\ell) = \kappa(\ell) \left[1 - \chi_\kappa \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (2)$$

This modifies the birth and death thresholds of topological features:

$$b_{\text{RDCC}} = b[1 - \chi_b(\ell/\ell_{\text{rel}})^\alpha], \quad (3)$$

$$d_{\text{RDCC}} = d[1 - \chi_d(\ell/\ell_{\text{rel}})^\alpha]. \quad (4)$$

The ensemble of RDCC diagrams becomes

$$\{D_{i,\text{RDCC}}\} = \{(b_{i,\text{RDCC}}, d_{i,\text{RDCC}})\}. \quad (5)$$

4 Fréchet Means in RDCC

The RDCC Fréchet mean is

$$D_{\text{RDCC}}^* = \arg \min_D \sum_i W_p(D, D_{i,\text{RDCC}})^p. \quad (6)$$

To first order in the relaxation parameter,

$$D_{\text{RDCC}}^* = D^* \left[1 - \chi_F \left(\frac{\ell}{\ell_{\text{rel}}} \right)^\alpha \right]. \quad (7)$$

This modification reflects the relaxation-driven shift in ensemble topology.

5 Ensemble-Level Signatures of RDCC

The relaxation mechanism enhances the coherence of large-scale modes. Fréchet means therefore exhibit:

- a shift toward longer-lived large-scale features,
- suppression of short-lived small-scale features,
- reduced ensemble variance at large scales,
- enhanced variance at small scales,
- a characteristic turnover near ℓ_{rel} ,
- modified clustering of diagrams in Fréchet space.

These signatures provide a direct ensemble-level probe of the RDCC gravitational field.

6 Conclusion

Fréchet means of persistence diagrams in the Relaxation-Driven Cyclic Cosmology reflect the scale-dependent evolution of the gravitational potential and the nonlinear matter distribution. The relaxation mechanism modifies the ensemble topology in a scale-dependent manner, producing distinctive signatures in weak-lensing surveys. These effects provide a direct observational window into the relaxation scale and the temporal structure of the RDCC gravitational field. The present Companion establishes the theoretical foundation for future studies of topological ensemble statistics, nonlinear structure, and the interplay between light and gravity in RDCC.

References

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